

Aptitude, 2018 EC, entrance exam, version 3

Grade 12 · Aptitude · 2018 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. COURAGEOUS is to TIMID as STUNNING is to _____ (1)
 A. Graceful B. Hideous
 C. Frightful D. Slothful
2. PROLOGUE is to INTRODUCTION as EPILOGUE is to _____ (1)
 A. Beginning B. Ending
 C. Chapter D. Conclusion
3. BOOK is to LIBRARY as ARTIFACT is to _____ (1)
 A. Church B. Gallery
 C. Museum D. Warehouse
4. PRECAUTION is to ACCIDENT as CLEANLINESS is to _____ (1)
 A. Garbage B. Filthy
 C. Treatment D. Disease
5. Which word is most nearly similar in meaning to DILIGENT? (1)
 A. Hardworking B. Lazy
 C. Careless D. Negligent
6. Which word is most nearly similar in meaning to EXORBITANT? (1)
 A. Moderate B. Excessive
 C. Reasonable D. Cheap
7. Which word is most nearly similar in meaning to AUTONOMOUS? (1)
 A. Reliant B. Intellectual
 C. Worldly D. Independent
8. Which word is NEARLY OPPOSITE in meaning to LIGHTEN? (1)
 A. Transpose B. Fortify
 C. Exacerbate D. Rectify
9. Which word is NEARLY OPPOSITE in meaning to PROFOUND? (1)
 A. Deep B. Superficial
 C. Serious D. Thoughtful
10. Which word is NEARLY OPPOSITE in meaning to DARING? (1)
 A. Valiant B. Cowardly
 C. Nervous D. Unpleasant

-
11. Which word cannot be categorized under the same group? Reader, Publisher, Reporter, Writer (1)
- | | |
|-------------|--------------|
| A. Reader | B. Publisher |
| C. Reporter | D. Writer |
-
12. Which word cannot be categorized under the same group? Violin, Drum, Piano, Guitar (1)
- | | |
|-----------|-----------|
| A. Violin | B. Drum |
| C. Piano | D. Guitar |
-
13. Which word cannot be categorized under the same group? Albert Einstein, William Shakespeare, Chinua Achebe, Wole Soyinka (1)
- | | |
|--------------------|------------------------|
| A. Albert Einstein | B. William Shakespeare |
| C. Chinua Achebe | D. Wole Soyinka |
-
14. The teacher admired the student's _____ approach to research, showing diligence and critical thinking. (1)
- | | |
|--------------|-----------------|
| A. Scholarly | B. Uncaring |
| C. Negligent | D. Conventional |
-
15. The researcher's conclusions were based on _____ evidence collected over many years. (1)
- | | |
|--------------|-----------------|
| A. Erroneous | B. Substantial |
| C. Imaginary | D. Questionable |
-
16. The diplomat handled the sensitive negotiations with _____, balancing multiple interests carefully. (1)
- | | |
|---------------|-----------------|
| A. Clumsiness | B. Partiality |
| C. Finesse | D. Recklessness |
-
17. The teacher praised the student for her _____ response, which showed deep understanding of the topic. (1)
- | | |
|---------------|---------------|
| A. Biased | B. Insightful |
| C. Reflective | D. Inflated |
-
18. The artist's style is _____, blending traditional and modern techniques seamlessly. (1)
- | | |
|-------------|---------------|
| A. Uniform | B. Monotonous |
| C. Eclectic | D. Derivative |
-
19. Five seminars A, B, C, D, and E are scheduled from Monday to Friday, one each day. Conditions: A is scheduled before C. D is scheduled after B. E is scheduled on Wednesday. A is placed before Friday immediately after E. Which of the following is the correct order of seminars? (1)
- | | |
|------------------|------------------|
| A. B, D, E, A, C | B. D, B, E, A, C |
| C. B, E, A, D, C | D. E, B, D, A, C |
-
20. Based on the same seminar scheduling conditions, which of the following statements must be true? (1)
- | | |
|-----------------------------|--------------------------------|
| A. D is scheduled on Friday | B. C is scheduled on Thursday |
| C. B is scheduled on Monday | D. A is scheduled on Wednesday |
-
21. Based on the same seminar scheduling conditions, which of the following is not possible? (1)
- | | |
|-------------------|------------------|
| A. A on Thursday | B. C on Friday |
| C. E on Wednesday | D. D on Thursday |

-
- 22.** Ali said, 'The man is the brother of my mother's only daughter.' How is the man related to Ali? (1)
- A. Nephew
B. Brother
C. Uncle
D. Cousin
-
- 23.** Statement: 'No student who fails in mathematics can participate in the science fair.' Conclusion: I. If a student participates in the science fair, they have passed mathematics. II. If a student fails in mathematics, they cannot participate. III. All students who pass mathematics will participate in the science fair. (1)
- A. Only conclusion I follows
B. Only conclusion II follows
C. Both conclusion I and II follow
D. All conclusions follow
-
- 24.** Statement: 'No person without a ticket is allowed inside the concert hall.' Conclusion: I. If a person is inside the concert hall, they must have a ticket. II. If a person has a ticket, they must enter. III. All ticket holders enter the concert hall. (1)
- A. Only conclusion I follows
B. Only conclusion II follows
C. Both conclusion I and II follow
D. All conclusions follow
-
- 25.** *Vision is a critical trait in leaders and innovators, providing a clear sense of direction and purpose. Those with vision anticipate future challenges, set ambitious goals, and inspire others to pursue shared objectives. Visionary thinking often drives breakthroughs in technology, business, and social reform. Yet, vision alone is insufficient. Successful implementation requires strategic planning, collaboration, and adaptability to unforeseen obstacles. Overconfidence in visionary ideas can result in impractical projects or failed initiatives if execution is neglected. Moreover, vision must align with societal needs. Leaders who pursue personal ambition without considering ethical or social consequences risk alienating supporters and undermining long-term success. Effective vision balances ambition with responsibility.* (1)
- The main idea of the passage is:
- A. Vision is unnecessary for leadership.
B. Ambitious goals are always successful.
C. Leaders should prioritize personal ambition above all.
D. Vision must be coupled with strategy and ethics to be effective.
-
- 26.** *Vision is a critical trait in leaders and innovators, providing a clear sense of direction and purpose. Those with vision anticipate future challenges, set ambitious goals, and inspire others to pursue shared objectives. Visionary thinking often drives breakthroughs in technology, business, and social reform. Yet, vision alone is insufficient. Successful implementation requires strategic planning, collaboration, and adaptability to unforeseen obstacles. Overconfidence in visionary ideas can result in impractical projects or failed initiatives if execution is neglected. Moreover, vision must align with societal needs. Leaders who pursue personal ambition without considering ethical or social consequences risk alienating supporters and undermining long-term success. Effective vision balances ambition with responsibility.* (1)
- Which of the following best describes a risk of visionary thinking?
- A. Encouraging social reform
B. Inspiring others to pursue goals
C. Failure from bad implementation
D. Predicting future challenges accurately

-
27. *Vision is a critical trait in leaders and innovators, providing a clear sense of direction and purpose. Those with vision anticipate future challenges, set ambitious goals, and inspire others to pursue shared objectives. Visionary thinking often drives breakthroughs in technology, business, and social reform. Yet, vision alone is insufficient. Successful implementation requires strategic planning, collaboration, and adaptability to unforeseen obstacles. Overconfidence in visionary ideas can result in impractical projects or failed initiatives if execution is neglected. Moreover, vision must align with societal needs. Leaders who pursue personal ambition without considering ethical or social consequences risk alienating supporters and undermining long-term success. Effective vision balances ambition with responsibility.* (1)

'Alienating' in paragraph 3 most nearly means:

- | | |
|---------------|---------------|
| A. Uniting | B. Isolating |
| C. Motivating | D. Supporting |
-

28. *Vision is a critical trait in leaders and innovators, providing a clear sense of direction and purpose. Those with vision anticipate future challenges, set ambitious goals, and inspire others to pursue shared objectives. Visionary thinking often drives breakthroughs in technology, business, and social reform. Yet, vision alone is insufficient. Successful implementation requires strategic planning, collaboration, and adaptability to unforeseen obstacles. Overconfidence in visionary ideas can result in impractical projects or failed initiatives if execution is neglected. Moreover, vision must align with societal needs. Leaders who pursue personal ambition without considering ethical or social consequences risk alienating supporters and undermining long-term success. Effective vision balances ambition with responsibility.* (1)

Which of the following is least consistent with the author's argument?

- A. Strategic planning strengthens visionary leadership.
 - B. Leaders who ignore societal impact risk losing support.
 - C. Ambition should be pursued regardless of ethical implications.
 - D. Adaptability is necessary for overcoming unforeseen obstacles.
-

29. *Climate change presents one of the most urgent challenges of the 21st century. Rising global temperatures, melting ice caps, and extreme weather events threaten ecosystems, economies and human livelihoods. Scientific consensus indicates that anthropogenic greenhouse gas emissions are a primary driver of these changes. Efforts to mitigate climate change include transitioning to renewable energy, enhancing energy efficiency, and implementing sustainable agricultural practices. While progress has been made in some regions, global cooperation remains inconsistent, and political or economic interests often impede meaningful action. Adaptation is equally important. Communities must develop resilience to the impacts of climate change, from flood defenses to drought-resistant crops. The interplay between mitigation and adaptation determines humanity's ability to navigate an increasingly unstable climate system.* (1)

The primary purpose of the passage is:

- A. To deny the existence of climate change
- B. To explain why adaptation is unnecessary
- C. To focus solely on renewable energy technologies
- D. To highlight challenges and solutions related to climate change

-
- 30.** *Climate change presents one of the most urgent challenges of the 21st century. Rising global temperatures, melting ice caps, and extreme weather events threaten ecosystems, economies and human livelihoods. Scientific consensus indicates that anthropogenic greenhouse gas emissions are a primary driver of these changes. Efforts to mitigate climate change include transitioning to renewable energy, enhancing energy efficiency, and implementing sustainable agricultural practices. While progress has been made in some regions, global cooperation remains inconsistent, and political or economic interests often impede meaningful action. Adaptation is equally important. Communities must develop resilience to the impacts of climate change, from flood defenses to drought-resistant crops. The interplay between mitigation and adaptation determines humanity's ability to navigate an increasingly unstable climate system.* (1)
- Which of the following is an alleviation strategy mentioned in the passage?

- A. Practicing harmful agriculture
 - B. Using energy more efficiently
 - C. Relying on non-renewable energy
 - D. Overlooking greenhouse gas emissions
-

- 31.** *Climate change presents one of the most urgent challenges of the 21st century. Rising global temperatures, melting ice caps, and extreme weather events threaten ecosystems, economies and human livelihoods. Scientific consensus indicates that anthropogenic greenhouse gas emissions are a primary driver of these changes. Efforts to mitigate climate change include transitioning to renewable energy, enhancing energy efficiency, and implementing sustainable agricultural practices. While progress has been made in some regions, global cooperation remains inconsistent, and political or economic interests often impede meaningful action. Adaptation is equally important. Communities must develop resilience to the impacts of climate change, from flood defenses to drought-resistant crops. The interplay between mitigation and adaptation determines humanity's ability to navigate an increasingly unstable climate system.* (1)
- 'Resilience' in paragraph 3 most nearly means:

- A. Ignorance
 - B. Vulnerability
 - C. Resistance to change
 - D. Ability to recover and adapt to challenges
-

- 32.** *Climate change presents one of the most urgent challenges of the 21st century. Rising global temperatures, melting ice caps, and extreme weather events threaten ecosystems, economies and human livelihoods. Scientific consensus indicates that anthropogenic greenhouse gas emissions are a primary driver of these changes. Efforts to mitigate climate change include transitioning to renewable energy, enhancing energy efficiency, and implementing sustainable agricultural practices. While progress has been made in some regions, global cooperation remains inconsistent, and political or economic interests often impede meaningful action. Adaptation is equally important. Communities must develop resilience to the impacts of climate change, from flood defenses to drought-resistant crops. The interplay between mitigation and adaptation determines humanity's ability to navigate an increasingly unstable climate system.* (1)
- Why does the author emphasize both mitigation and adaptation?

- A. Because both focus only on short-term effects
 - B. Because both are needed to manage climate impacts
 - C. Because both try to solve climate change separately
 - D. Because both assume reducing emissions alone is enough
-

- 33.** If the numerator and denominator of a proper fraction are increased by the same quantity, the resulting fraction is: (1)
- A. Always greater than the original fraction
 - B. Always less than the original fraction
 - C. Always equal to the original fraction
 - D. Cannot be determined

34. What number should replace X to continue the sequence 2, 6, 14, 30, 62, x? (1)

A. 122
B. 124
C. 126
D. 128

35. Last year, Heran saved 10% of her annual earnings. This year she earned 5% more than last year and she saved 12% of her annual earnings. The amount saved this year is what percent of the amount saved last year? (1)

A. 123%
B. 124%
C. 125%
D. 126%

36. A telecom company provides an internet package that charges a service fee of Birr 700 per month plus Birr 0.75 per megabyte of data used. If a man uses 5000 megabytes of data in one month, how many birr will he pay to the telecom company for that month? (1)

A. 4000
B. 1500
C. 4450
D. 3750

37. Average age of seven persons in a group is 30 years. The average age of five persons of this group is 31 years. What is the average age of the other two persons in the group? (1)

A. 55 years
B. 26 years
C. 27.5 years
D. 15 years

38. Three partners, Gemechu, Gurmu and Lemi start a business. Twice Gemechu's capital is equal to three times Gurmu's capital and Gurmu's capital is four times Lemi's capital. Out of a total profit of Birr 16500, what is the share of Gurmu? (1)

A. Birr 1500
B. Birr 6000
C. Birr 4000
D. Birr 6600

39. Two pipes, A and B, can fill a tank in 20 minutes and 30 minutes respectively. If both pipes are used together, how long will it take to fill the tank? (1)

A. 12
B. 15
C. 25
D. 5

40. Which of the following fractions is the largest? $\frac{7}{8}$, $\frac{13}{16}$, $\frac{31}{40}$, $\frac{63}{80}$ (1)

A. $\frac{7}{8}$
B. $\frac{13}{16}$
C. $\frac{31}{40}$
D. $\frac{63}{80}$

41. The sum of the squares of three numbers is 138, and the sum of their products taken two at a time is 131. What is the sum of the three numbers? (1)

A. 400
B. 20
C. 262
D. 40

42. An observer 1.6m tall is $20\sqrt{3}$ m away from a tower. The angle of elevation from his eye to the top of the tower is 30° . The height of the tower is: (1)

A. 21.6 m
B. 23.2 m
C. 24.7 m
D. 20 m

43. For any real number a, which of the following is always true? (1)

A. $(a-1)^3 / (a-1) = a^2 - 2a + 1$
B. $4 + (2a)^2 / (a^2 + 1) = 4$
C. $a^{(-1/2)} = 1/\sqrt{a}$
D. $a^0 = 1$

-
- 44.** A shop owner selling x pairs of shoes per day has a profit (in birr) expressed by the equation $p(x) = -x^2 + 25x - 10$. If the owner needs to obtain a profit of more than 140 birr per day, then the number of pairs of shoes the shop owner makes should range between: (1)
- A. 5 and 10
B. 10 and 15
C. 5 and 15
D. 12 and 15
-
- 45.** Compare the two quantities: Quantity A = $(-51)^{16}$, Quantity B = $(-57)^{17}$ (1)
- A. Quantity A is less than quantity B
B. Quantity A is equal to quantity B
C. Quantity A is greater than quantity B
D. The relationship cannot be determined
-
- 46.** Three candidates contested an election and received 1136, 7636, and 11628 votes respectively. What percentage of the total votes did the winning candidate receive? (1)
- A. 57%
B. 60%
C. 65%
D. 90%
-
- 47.** Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5? (1)
- A. $\frac{1}{2}$
B. $\frac{2}{5}$
C. $\frac{8}{15}$
D. $\frac{9}{20}$
-
- 48.** A mixture contains 120 litres of mango and avocado juice in the ratio 2:3. If the ratio is to be changed to 3:2, how many litres of avocado juice must be removed? (1)
- A. 40
B. 48
C. 78
D. 20
-
- 49.** If $f(x)$ and $g(x)$ are logarithmic functions whose bases are reciprocals of each other, then which one of the following statements is true? (1)
- A. They have reciprocal values for all x .
B. There are some equal values of x for which both $f(x)$ and $g(x)$ are positive.
C. They have opposite signs for all $x \neq 1$.
D. They can never have equal values for any x .
-
- 50.** How many 4-letter words, with or without meaning, can be formed out of the letters of the word 'LOGARITHMS', if repetition of letters is not allowed? (1)
- A. 40
B. 400
C. 2540
D. 5040
-
- 51.** $[(469+174)^2 - (469-174)^2] / (469 \times 174)$ is equal to: (1)
- A. 2
B. 4
C. 295
D. 643
-

— END OF EXAMINATION —